## Ilya Meerovich Sobol'
(On his 70th birthday)

On August 15, 1996 professor Ilya M. Sobol', an outstanding specialist in applied mathematics and mathematical modelling and the Principal Researcher of the Institute for Mathematical Modelling of the Russian Academy of Sciences, was 70.

Ilya Meerovich was born in Panevežys (Lithuania). During the years 1941 – 1943 he lived in Izhevsk where he finished with honorous the high school. Since 1944 he lives in Moscow.

In 1948 he graduated with honours from the Moscow State Lomonosov University (Faculty of Mechanics and Mathematics) where he attended lectures given by famous mathematicians like A.N.Kolmogorov, I.G.Petrovsky, M.V.Keldysh, A.Ya.Khinchin, I.M.Gelfand, P.S.Alexandrov, D.E.Menshov, V.V.Niemytzki. At this time his gift for mathematics became already clear. Working in the seminar on qualitative theory of ordinary differential equations under V.V.Stepanov and V.V.Niemytzki, I.M.Sobol' wrote a paper on the *asymptotic behaviour of solutions of linear differential equations* that was published in the Doklady (Comptes Rendus) of the USSR Academy of Sciences. His papers on the *Riccati equation* were published also.

These were the first papers published by I. M. Sobol'. Now his list of scientific publications consists of more than 150 items, including books, articles, papers presented at international conferences.

In 1949, he started his work at a new Laboratory N 8 of the Complex Geophysical Expedition (affiliated to the Geophysical Institute of the USSR Academy of Sciences) that has been recently organized by the Government. This laboratory was later included into the Institute of Applied Mathematics of the USSR Acad. Sci. His participation in the work was rather active. For his part in computing processes of nuclear and thermonuclear explosions he received government's awards: a medal "For Valiant Labor" and an order "The Badge of Honour".

In 1954, I.M.Sobol' started investigations of *Monte Carlo methods* and most of his forthcoming works are directly or indirectly connected with this subject. Nevertheless, among the exceptions is a paper on computing temperature waves (co-authored with A.A.Samarski, 1963) where for the first time *a blow-up solution* featuring heat localization was found.

An original *pseudorandom number generator*, the first in our country, was published in 1958. In the note "Pseudorandom numbers for the computer Strela" the possibility of a deterministic approach to Monte Carlo computations of mathematical expectations was discussed that later received the name *quasi-Monte Carlo*. In 1962, I.M.Sobol' became one of the authors of the book *"The Method of Statistical Trials (Monte Carlo Method)"* that was translated into several foreign languages.

Investigations in the *theory of uniform distribution* started having in mind the construction of quasi-random points. In 1959, the first dissertation (Candidate of Phys.-Math. Sciences) was defended, and in 1966 a new type of sequences of multidimensional points (called $LP_\tau - sequences$) was introduced. Their uniformity is asymptotically optimal. In 1976, he succeeded in constructing $LP_\tau$-sequences with additional uniformity properties. Even now, these sequences remain the best from the practical point of view; they are used in various computations and often called *"Sobol sequences"*.

In 1969, a monograph *"Multidimensional Quadrature Formulas and Haar Functions"* appeared, and in 1972 his second dissertation (Doctor of Phys.-Math. Sciences) was

defended: *"The Method of Haar Series in the Theory of Quadrature Formulas"*. In 1973 a book *"Numerical Monte Carlo Methods"* appeared where one can find concepts introduced by the author: *constructive dimension of Monte Carlo algorithms, statistical vector weights*, etc.

Since 1977 Ilya Meerovich began to study problems in *multicriteria decision making*. He is one of the authors of an efficient optimization method where points of $LP_\tau$-sequences are used as trial points in the parameter space. Scientists from the Institute for Machine Research (USSR Acad. Sci.) and from other institutes often apply the method for optimum design of different mechanisms and machines. In 1981 a book *"Selection of Optimal Parameters in Problems with Several Criteria"* was published.

I.M.Sobol's contribution to astrophysics is significant. In particular, for about ten years (1976 – 1986) he took part in modelling of X-ray and gamma-ray spectra of compact relativistic objects. He developed Monte Carlo methods for computing comptonized radiation and these methods allowed to model spectra of black holes, neutron stars, supernovae, etc.

A series of papers (1982 – 1994) contain investigations of errors in computational algorithms for *nonisotropic classes of multivariable functions*. These results allowed to explain why algorithms that are theoretically optimal for classes of functions quite often perform poorly in practice.

In 1990, he introduced *global sensitivity indices* that are now a widely approved tool for studying nonlinear multivariable functions defined not by formulas but rather by a program. A Monte Carlo method has been constructed that makes possible the estimation of such indices in most general situations.

A small book *"The Monte Carlo Method"* (1968, Ser. "Popular Lectures in Mathematics") must be mentioned: a lot of people made their first acquaintance with the Monte Carlo method through this book. It was several times revised and complemented, and appeared in Russian, German, English and several other languages including Spanish and Persian. The newest version of this book appeared in the USA: *"A Primer for the Monte Carlo Method"*, CRC Press, 1994.

Ilya Meerovich is a talented lecturer, since 1969 he is teaching at the Moscow Engineering Physics Institute (Technical University) and is a Professor of the Chair of Applied Mathematical Physics. More than 20 students defended their thesis under his supervision, there are Candidates and Doctors of Sciences among his graduates.

I.M.Sobol' is a member of the Editorial Board of the international journal *"Monte Carlo Methods and Applications"* and an active member of the New York Academy of Sciences.

Ilya Meerovich is an extremely modest and very diligent person, his creative abilities are great. He is active in different fields of mathematics, he suggests original approaches and finds new solutions, he participates in international conferences. His works are known to the international scientific community and are often cited in various countries. He is full with new ideas and interesting projects. We wish him good health, new creative successes, and luck in all his enterprises.

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